

## Part 2a: Static Game Theory

**Instructor:** Oleg Baranov

Microeconomics 2 (*Module 1*)

# Suggested Readings for this part

## Readings in Class Textbooks:

- ▶ **Varian, 8th edition:**

- Ch 28: Game Theory
- Ch 29: Game Applications

- ▶ **Carlton and Perloff:**

- Ch 6: Oligopoly

# Motivation

- ▶ Our focus has been on markets dominated by a **single** firm.
  - **Goal:** how to create and extract surplus from consumers.
  - **Choices:** pricing (linear and non-linear), consumer pricing, product differentiation, product design (vertical differentiation).
- ▶ But the actual monopoly markets are rare in practice.
- ▶ Instead, many markets are dominated by a small number of large firms. This situation is called **oligopoly**.

# Oligopoly

**Many markets are dominated by a **small** number of **large** firms:**

- ▶ Soft drinks — Coke and Pepsi.
- ▶ Smartphones — Apple, Samsung, Google, ...
- ▶ Each firm's action affects others, and prompts reactions.
- ▶ Firms should take these reactions into account when formulating strategies for pricing, output, new products, ....

**Thus, strategic component (interaction among firms) is important here.**

# Oligopoly

- ▶ A **competitive firm** has no market power (price-taking)

$$\max_{q \geq 0} p \cdot q - C(q)$$

- ▶ A **monopoly** has all market power (sets any price/quantity)

$$\max_{q \geq 0} p(q) \cdot q - C(q)$$

- ▶ With **oligopoly**, there are several big firms. Each one maximizes its own profit by
  - setting prices/quantities
  - while **accounting for choices of others** since they also affect its profit
- ▶ We have to use **game theory** – a mathematical tool to analyze strategic interactions and conflicts of interest.

# **Static Game Theory**

## **(One-period games)**

## Example of a static game

Competition between **Aeroflot** and **S7** on the route between Moscow and Saint-Petersburg.

- ▶ **Possible actions:** time of the flight — {Morning, Evening}.
- ▶ **Profits:**
  - Demand for morning = 80
  - Demand for evening =  $A$  ( $A \geq 80$ )
  - Each airline can serve up to 100 passengers at a time.
  - Constant marginal cost  $c = 0$  (*for simplicity*)
  - Airfare (price of a ticket) fixed at  $p = 1$ .
  - If they fly at the same time, they split the market.
  - If they fly at different times, each gets the whole market.

## Quick Preview: Large $A$

- Suppose that  $A \geq 200$

|                 |         | <b>S7</b> |          |
|-----------------|---------|-----------|----------|
|                 |         | Morning   | Evening  |
| <b>Aeroflot</b> | Morning | 40, 40    | 80, 100  |
|                 | Evening | 100, 80   | 100, 100 |



## Quick Preview: Large A

- Suppose that  $A \geq 200$

|                 |         | <b>S7</b> |          |
|-----------------|---------|-----------|----------|
|                 |         | Morning   | Evening  |
| <b>Aeroflot</b> | Morning | 40, 40    | 80, 100  |
|                 | Evening | 100, 80   | 100, 100 |

- One prediction
- For each airline, **Morning** is strictly dominated by **Evening**

## Quick Preview: Small A

- Suppose that  $A = 120$

|                 |         | <b>S7</b> |         |
|-----------------|---------|-----------|---------|
|                 |         | Morning   | Evening |
| <b>Aeroflot</b> | Morning | 40, 40    | 80, 100 |
|                 | Evening | 100, 80   | 60, 60  |

## Quick Preview: Small A

- ▶ Suppose that  $A = 120$

|                 |         | <b>S7</b> |         |
|-----------------|---------|-----------|---------|
|                 |         | Morning   | Evening |
| <b>Aeroflot</b> | Morning | 40, 40    | 80, 100 |
|                 | Evening | 100, 80   | 60, 60  |

- ▶ Two predictions
- ▶ Later we will say that the game has two Nash equilibria
- ▶ This game is known as “coordination game”.

## Quick Preview: Intermediate A

- Suppose that  $A = 160$

|                 |         | <b>S7</b> |         |
|-----------------|---------|-----------|---------|
|                 |         | Morning   | Evening |
| <b>Aeroflot</b> | Morning | 40, 40    | 80, 100 |
|                 | Evening | 100, 80   | 80, 80  |

## Quick Preview: Intermediate A

- ▶ Suppose that  $A = 160$

|                 |         | <b>S7</b> |         |
|-----------------|---------|-----------|---------|
|                 |         | Morning   | Evening |
| <b>Aeroflot</b> | Morning | 40, 40    | 80, 100 |
|                 | Evening | 100, 80   | 80, 80  |

- ▶ Three predictions
- ▶ For each airline, **Morning** is weakly dominated by **Evening**

# Definition of Simultaneous Game

- ▶ Set of **players**  $N$
- ▶ For each player from  $N$ , set of **actions/pure strategies** available to him/her:
  - $S_i$  - set of all possible actions for player  $i$
  - $s_i$  - a particular play of player  $i$
  - $s = (s_1, \dots, s_N)$  a particular play of all players (strategy profile)
  - $s_{-i}$  a particular play of all players except player  $i$
- ▶ **Payoffs/utilities/profits**: A payoff of player  $i$  at strategy profile  $s$  (particular play of the game)

$$u_i(s_1, s_2, \dots, s_N)$$

$$u_i(s)$$

$$u_i(s_i, s_{-i})$$

## Example 1

|          |   | Player 2 |      |
|----------|---|----------|------|
|          |   | L        | R    |
| Player 1 | T | 1, 5     | 4, 6 |
|          | M | 3, 11    | 4, 0 |
|          | B | 0, 0     | 0, 2 |

## Example 1

|          |   | Player 2 |      |
|----------|---|----------|------|
|          |   | L        | R    |
| Player 1 | T | 1, 5     | 4, 6 |
|          | M | 3, 11    | 4, 0 |
|          | B | 0, 0     | 0, 2 |

**Players:**

1,2

**Strategy Sets:**

$S_1 = \{T, M, B\}$

$S_2 = \{L, R\}$

**Payoffs:**

$u_1(T, L) = 1$

$u_2(T, L) = 5$

...



## Example 2: “Rock, Paper, Scissors” Game

|          |          | Player II |       |          |
|----------|----------|-----------|-------|----------|
|          |          | Rock      | Paper | Scissors |
| Player I | Rock     | 0, 0      | -1, 1 | 1, -1    |
|          | Paper    | 1, -1     | 0, 0  | -1, 1    |
|          | Scissors | -1, 1     | 1, -1 | 0, 0     |

**Play this game on the Smart LMS for the next time.**

# **Solution Concepts**

## **Dominance Principle**

# First Solution Concept – *Dominance*

## Definition: Strict Dominance

For player  $i$ , strategy  $x$  **strictly dominates** some other strategy  $y$  if strategy  $x$  generates a **strictly higher** payoff for player  $i$  than strategy  $y$  **no matter what other players are playing**, i.e.,

$$u_i(x, s_{-i}) > u_i(y, s_{-i}) \quad \text{for all} \quad s_{-i}.$$

*Alternative language:*

- ▶ If  $x$  **strictly dominates**  $y$ , then  $y$  is **strictly dominated** by  $x$ .
- ▶ If  $x$  strictly dominates ALL other strategies in  $S_i$ , then  $x$  is **strictly dominant** strategy for player  $i$ .

## Class Example 1

|          |   | Player 2 |      |
|----------|---|----------|------|
|          |   | L        | R    |
| Player 1 | T | 1, 5     | 4, 6 |
|          | M | 3, 11    | 4, 0 |
|          | B | 0, 0     | 0, 2 |

# First Solution Concept – *Dominance*

## Definition: Weak Dominance

For player  $i$ , strategy  $x$  **weakly dominates** some other strategy  $y$  if strategy  $x$  generates a **weakly higher** payoff for player  $i$  than strategy  $y$  no matter what other players are playing, i.e.,

$$u_i(x, s_{-i}) \geq u_i(y, s_{-i}) \quad \text{for all } s_{-i}$$

$$u_i(x, s_{-i}) > u_i(y, s_{-i}) \quad \text{for some } s_{-i} \quad (\text{at least one } s_{-i})$$

*Alternative language:*

- ▶ If  $x$  **weakly dominates**  $y$ , then  $y$  is **weakly dominated** by  $x$
- ▶ If  $x$  weakly dominates ALL other strategies in  $S_i$ , then  $x$  is **weakly dominant** strategy for player  $i$ .

## First Solution Concept – Dominance (strict and weak)

Is it a good solution concept? Do we believe in its prediction?

- ▶ Dominance makes sense

*There are some empirical evidence that people do not play dominated strategies*

- ▶ But it is not powerful enough to solve majority of interesting games (no clear predictions).

*Example:* A “coordination” game.

# Grade Game

**Each student have to pick between two options: A or B.**

I will randomly pair your form with one other form. Neither you nor your pair will ever know with whom you were paired. We will be using the **HSE 10-point grading system**. Here is how grades may be assigned for this class (hypothetically).

- ▶ If you put **A** and your pair puts **B**, then you will get **9**, and your pair will get **4** .
- ▶ If both of you put **A**, then you both will get grade **5**.
- ▶ If you put **B** and your pair puts **A**, then you will get grade **4**, and your pair will get grade **9**.
- ▶ If both of you put **B**, then you will both get grade **7**.

# Guess $2/3$ of the Average Game

## Rules of the game:

Each of you have to choose an integer between 1 and 100 in order to guess " $2/3$  of the average of the responses given by all students in the class". Each student who guesses the integer closest to the  $2/3$  of the average of all the responses, wins.

## Example: (with only 3 students)

|                      |                      |
|----------------------|----------------------|
| Choices made:        | 42, 13, 80           |
| The average:         | 45                   |
| $2/3$ of the Average | 30                   |
| <b>The winner</b>    | Student who wrote 42 |



# **Solution Concepts**

## **Iterative Dominance Principle**

## Guess 2/3 of the Average Game: Solving

| Strategies:     | Why not to play?                                                                                        | Requirements                                                          |
|-----------------|---------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| 68, 69,..., 100 | weakly dominated by 67                                                                                  | my rationality (R)                                                    |
| 45,..., 67      | not weakly dominated<br>in the original game<br>but weakly dominated by 44<br>once we eliminated 68-100 | my rationality (R)<br>+<br>knowledge that others<br>are rational (KR) |
| 30,..., 44      | weak. dominated by 29 ...                                                                               | R, KR, KKR                                                            |
| 20,..., 29      | weak. dominated by 19                                                                                   | ...                                                                   |
| ...             | ...                                                                                                     | ...                                                                   |
| 2               | weak. dominated by 1                                                                                    | ...                                                                   |

**After multiple rounds of elimination, only 1 survives.**

**Let's see our own results**

# Second Solution Concept - Iterative Elimination of Strictly Dominated Strategies (IESDS)

## 1st round of elimination:

- ▶ Start with a game with actions sets:  $A_i$  for  $i = 1, \dots, N$ .
- ▶ For each player  $i$ , delete all actions which are strictly dominated.
- ▶ Obtain new game with smaller action sets:  
 $B_i = [\text{undominated actions of player } i] \subseteq A_i$  for  $i = 1, \dots, N$ .

## 2nd round of elimination:

- ▶ Start with a game with actions sets:  $B_i$  for  $i = 1, \dots, N$ .
- ▶ For each player  $i$ , delete all actions  $a_i$  which are strictly dominated.
- ▶ Obtain new game with smaller action sets:  
 $C_i = [\text{undominated actions of player } i] \subseteq B_i$  for  $i = 1, \dots, N$ .

**Repeat until no strictly dominated strategies left.**

## Class Example 2

|          |        | Player 2 |       |
|----------|--------|----------|-------|
|          |        | Left     | Right |
| Player 1 | Top    | 1, 2     | 4, 1  |
|          | Middle | 3, 2     | 2, 1  |
|          | Bottom | 2, 1     | 1, 3  |

# Properties of Iterated Elimination

## Strict Version (IESDS):

- ▶ If each player is left with one action, then we get a **unique** prediction. But this is very rare in practice.
- ▶ Order of elimination **does not matter** — a final set of actions is **always the same**.

## Weak Version (IEWDS):

- ▶ If each player is left with one action, then **one of possible** predictions.
- ▶ Order of elimination **does matter** — a final set of actions **can change**.

# *Iterated Elimination of Dominated Strategies (IEDS)*

Is it a good solution concept? Do we believe in its predictions?

1. Application of IEDS requires the knowledge of other players' payoffs (*unlike Dominance*).
2. Also requires the knowledge of other players' rationality (*unlike Dominance*).
3. Might require too many layers of rationality and lead to unrealistic predictions.

*Example:* “2/3 of the average” game

4. IEDS still can be powerless in many games (no clear predictions).

*Example:* A “coordination” game.

# **Solution Concepts**

## **Nash Equilibrium**



## Third Solution Concept - Nash Equilibrium

- ▶ **Main idea:** given a conjecture about the others, each player chooses her best action (playing the best response)
- ▶ **Definition:** Player  $i$ 's strategy  $x$  is a **best response** to the strategy profile  $s_{-i}$  of other players (denoted as  $BR_i(s_{-i})$ ) if

$$u_i(x, s_{-i}) \geq u_i(y, s_{-i}) \quad \text{for all} \quad y \in S_i$$

- ▶ **Nash equilibrium:**
  - each player forms a conjecture and plays its best response;
  - all conjectures turn up to be correct
- ▶ Formally, a **pure-strategy Nash equilibrium** is a strategy profile

$$s^* = (s_1^*, \dots, s_N^*) \quad \text{such that} \quad s_i^* \in BR_i(s_{-i}^*) \quad \forall i$$

## Class Example 3

|                 |        | <b>Player 2</b> |        |       |
|-----------------|--------|-----------------|--------|-------|
|                 |        | Left            | Center | Right |
| <b>Player 1</b> | Top    | 0, 4            | 4, 2   | 5, 2  |
|                 | Middle | 4, 0            | 3, 4   | 4, 2  |
|                 | Bottom | 2, 4            | 6, 5   | 3, 5  |

- Find all Nash equilibria

# Justification and Properties

**Main weakness:** This concept does not tell us how a player forms the **right conjecture**.

## Justifications of NE as a reasonable prediction:

- ▶ If agreed beforehand on this set of actions, then **nobody** has incentives to deviate.
- ▶ If one announces that she will play its part of NE, the others will also play the NE.
- ▶ Long-term interaction (play multiple times) where players learn what their opponents do.

## Properties of Nash equilibria:

- ▶ Nash equilibrium might **fail to exist** (no prediction).
- ▶ A game can have **several** Nash equilibria (multiple predictions)

# **Solution Concepts**

## **Relationship**

# Reminder: Three Solution Concepts for Static Games

## Dominated Strategies

*What not to play?*

## Iterative Elimination of Dominated Strategies

*What not to play deducing what others won't play?*

## Nash equilibrium

*Play your part of the mutual best response*

# Reminder: Three Solution Concepts for Static Games

## Dominated Strategies

*What not to play?*



## Iterative Elimination of Dominated Strategies

*What not to play deducing what others won't play?*

## Nash equilibrium

*Play your part of the mutual best response*

# Reminder: Three Solution Concepts for Static Games

## Dominated Strategies

*What not to play?*



## Iterative Elimination of Dominated Strategies

*What not to play deducing what others won't play?*



## Nash equilibrium

*Play your part of the mutual best response*

# Reminder: Three Solution Concepts for Static Games

## Dominated Strategies

*What not to play?*



## Iterative Elimination of Dominated Strategies

*What not to play deducing what others won't play?*



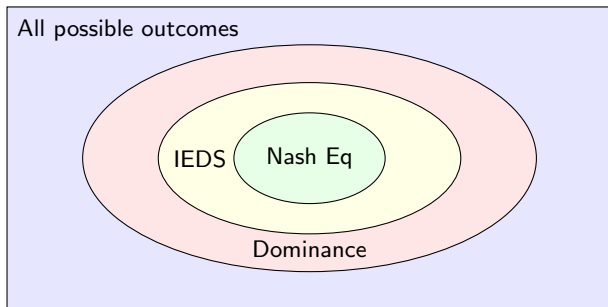
## Nash equilibrium

*Play your part of the mutual best response*





## Relationship among solution concepts

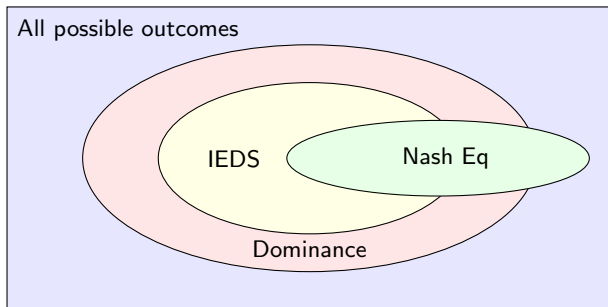


If **only strictly dominated strategies** are eliminated, then

**Dominance  $\supseteq$  IEDS  $\supseteq$  Nash equilibria**

*In other words, all Nash equilibria **survive** the iterative elimination of **strictly dominated strategies** (IESDS)*

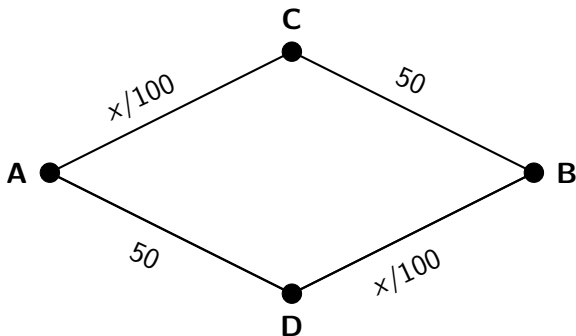
# Relationship among solution concepts



If both strictly and weakly dominated strategies are eliminated, then:

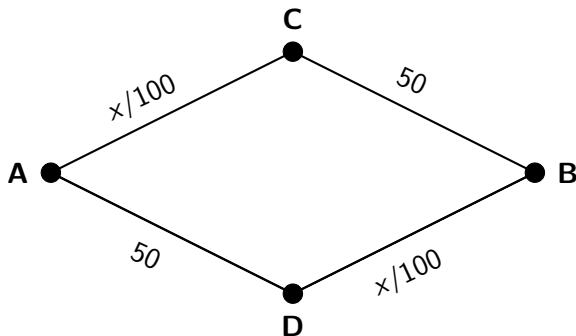
1.  $\text{Dominance} \supseteq \text{IEDS}$
2. IEDS contains at least one Nash equilibrium (if NE exists)
3. IEDS and Dominance are not necessarily contain all Nash equilibria

# Braess's Paradox



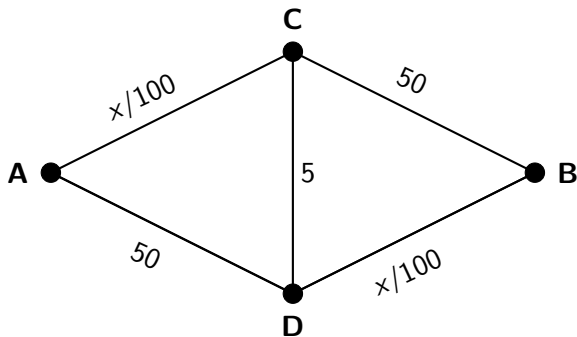
- ▶ 4000 cars need to get from A to B
- ▶ numbers on edges represent minutes needed to travel (AD and CB are congestion insensitive; AC and DB depend on congestion)

# Braess's Paradox



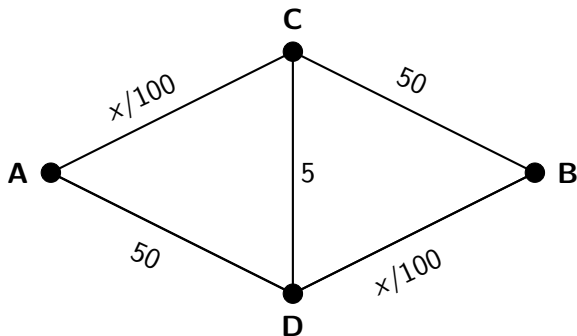
- ▶ 4000 cars need to get from A to B
- ▶ numbers on edges represent minutes needed to travel (AD and CB are congestion insensitive; AC and DB depend on congestion)
- ▶ **Nash equilibria:** any 2000 cars travel via C, and the other 2000 cars travel via D (**traveling time is 70 minutes**)

# Braess's Paradox



- Now C and D connected with a highway

# Braess's Paradox



- ▶ Now C and D connected with a highway
- ▶ **Unique Nash equilibrium:** all cars take ACDB
- ▶ **The equilibrium traveling time is 85 minutes**
- ▶ Building the CD connector is a bad idea

# Mixed Strategies

# "Rock, Paper, Scissors" Game

|          |          | Player II |       |          |
|----------|----------|-----------|-------|----------|
|          |          | Rock      | Paper | Scissors |
| Player I | Rock     | 0, 0      | -1, 1 | 1, -1    |
|          | Paper    | 1, -1     | 0, 0  | -1, 1    |
|          | Scissors | -1, 1     | 1, -1 | 0, 0     |



# “Rock, Paper, Scissors” Game

|          |          | Player II |       |          |
|----------|----------|-----------|-------|----------|
|          |          | Rock      | Paper | Scissors |
| Player I | Rock     | 0, 0      | -1, 1 | 1, -1    |
|          | Paper    | 1, -1     | 0, 0  | -1, 1    |
|          | Scissors | -1, 1     | 1, -1 | 0, 0     |

- ▶ No dominated strategies.
- ▶ Nash equilibrium in “pure” strategies does not exist.

**Why?** *If I play **Rock**, my opponent replies with **Paper**, but then my best response is to play **Scissors**, and so on ...*

**Mutual best response does not exist in this game!**

## Mixed strategies

A **mixed strategy** for player  $i \in N$  is a probability distribution  $p_i$  over  $S_i$ , that is

$$p_i(s_i) \geq 0 \quad \forall s_i \in S_i \quad \text{and} \quad \sum_{s_i \in S_i} p_i(s_i) = 1$$

where  $p_i(s_i)$  is **probability** that player  $i$  plays  $s_i$ . The set of all possible mixed strategies of player  $i$  is denoted  $\Delta_i$ .

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where  $p_i(s_i)$  is **probability** that player  $i$  plays  $s_i$ . The set of all possible mixed strategies of player  $i$  is denoted  $\Delta_i$ .

**Example:** Suppose  $S_i = \{Rock, Paper, Scissors\}$

- $p_i = (1/3, 1/3, 1/3)$  corresponds to player  $i$  choosing each action equally likely;
- $p_i = (0.5, 0.5, 0)$  corresponds to playing *Rock* and *Paper* with 50% probability each, and never playing *Scissors*
- $p_i = (0, 1, 0)$  corresponds to playing *Paper* with 100% probability

# Expected Payoffs

- ▶ Suppose that there are only **two** players.
- ▶ Player 1's **expected payoff** from a **pure strategy**  $s_i$  when another player is using a mixed strategy  $p_2$ :

$$EU_1(s_i, p_2) = \sum_{s_2 \in S_2} u_1(s_i, s_2) \cdot p_2(s_2)$$

- ▶ Player 1's **expected payoff** from a **mixed strategy**  $p_1$  when another player is using a mixed strategy  $p_2$ :

$$EU_1(p_1, p_2) = \sum_{s_i \in S_1} EU_1(s_i, p_2) \cdot p_1(s_i)$$

- ▶ We can define payoffs for Player 2 analogously.

## Class Example 4

**The class odds in Fall 2025 (104 responses):**

| <b>Rock</b> | <b>Paper</b> | <b>Scissors</b> |
|-------------|--------------|-----------------|
| 30%         | 51%          | 19%             |

## Class Example 4

**The class odds in Fall 2025 (104 responses):**

| Rock | Paper | Scissors |
|------|-------|----------|
| 30%  | 51%   | 19%      |

**Expected Payoff against  $p_2 = (0.30, 0.51, 0.19)$**

$$EU_1(Rock, p_2) = 0.30 [0] + 0.51 [-1] + 0.19 [1] = -0.32$$

$$EU_1(Paper, p_2) = 0.30 [1] + 0.51 [0] + 0.19 [-1] = 0.11$$

$$EU_1(Scissors, p_2) = 0.30 [-1] + 0.51 [1] + 0.19 [0] = 0.21$$

## Class Example 4

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**If player 1 plays  $p_1 = (1/3, 1/3, 1/3)$ , then**

$$EU_1(p_1, p_2) = \frac{1}{3}[-0.32] + \frac{1}{3}[0.11] + \frac{1}{3}[0.21] = 0$$

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**If player 1 plays  $p_1 = (1/3, 1/3, 1/3)$ , then**

$$EU_1(p_1, p_2) = \frac{1}{3}[-0.32] + \frac{1}{3}[0.11] + \frac{1}{3}[0.21] = 0$$

**But player 1 should play  $p_1 = (0, 0, 1)$  instead!**



# Rock, Paper, Scissors Game — HSE

| Class     | Rock | Paper | Scissors | Ranking     |
|-----------|------|-------|----------|-------------|
| Fall 2025 | 30%  | 51%   | 19%      | $S > P > R$ |
| Fall 2024 | 42%  | 30%   | 28%      | $P > R > S$ |
| Fall 2023 | 29%  | 42%   | 29%      | $S > P > R$ |
| Fall 2022 | 35%  | 40%   | 25%      | $P > S > R$ |

# Rock, Paper, Scissors Game — University of Colorado

| Class       | Rock  | Paper | Scissors | Ranking     |
|-------------|-------|-------|----------|-------------|
| Summer 2022 | 37%   | 39%   | 24%      | $P > S > R$ |
| Spring 2022 | 41%   | 33%   | 26%      | $P > R > S$ |
| Fall 2021   | 27%   | 44%   | 29%      | $S > P > R$ |
| Spring 2021 | 40%   | 43%   | 17%      | $P > S > R$ |
| Fall 2020   | 33%   | 41%   | 25%      | $P = S > R$ |
| Spring 2020 | 32.5% | 45.0% | 22.5%    | $S > P > R$ |
| Spring 2019 | 38.1% | 47.6% | 14.3%    | $P > S > R$ |
| Fall 2018   | 62.5% | 31.5% | 6%       | $P > R > S$ |

# Nash equilibrium in mixed strategies

- **Definition:** Player 1's mixed strategy  $p_1$  is a **best response** to Player 2 playing mixed strategy  $p_2$  (denoted as  $BR_1(p_2)$ ) if

$$EU_1(p_1, p_2) \geq EU_1(p', p_2) \quad \text{for all} \quad p' \in \Delta_i$$

- **Definition:** A **mixed strategy Nash equilibrium** is a profile  $p = (p_1, p_2)$  such that

$$p_1 \in BR_1(p_2) \quad \text{and} \quad p_2 \in BR_2(p_1)$$

## Nash Equilibrium in the 'Rock, Paper, Scissors' game

**If Player 2 plays  $p_2 = (1/3, 1/3, 1/3)$ , then**

$$EU_1(\text{Rock}, p_2) = \frac{1}{3} [0] + \frac{1}{3} [-1] + \frac{1}{3} [1] = 0$$

$$EU_1(\text{Paper}, p_2) = \frac{1}{3} [1] + \frac{1}{3} [0] + \frac{1}{3} [-1] = 0$$

$$EU_1(\text{Scissors}, p_2) = \frac{1}{3} [-1] + \frac{1}{3} [1] + \frac{1}{3} [0] = 0$$

**If Player 1 responds with  $p_1 = (1/3, 1/3, 1/3)$ , then**

$$EU_1(p_1, p_2) = \frac{1}{3}[0] + \frac{1}{3}[0] + \frac{1}{3}[0] = 0$$

**Can I do better than  $p_1$  above?**

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**Can I do better than  $p_1$  above?**

- ▶ No, Player 1 cannot do better.
- ▶ And Player 2 cannot do better against  $p_1$ .
- ▶ This is a **mixed strategy Nash equilibrium** for RPS game. **So what is the game theory actually predicts here?**

## How to play in *Rock, Paper, Scissors*?

- ▶ **An obvious answer:** If you somehow know the odds, just play **best response**.
- ▶ If you do not know the odds, there is no winning strategy (*the existence of such strategy is illogical*)
- ▶ But you can always eliminate any advantage of your opponent by **randomizing with equal probability** (*being completely unpredictable*)
- ▶ We cannot randomize in our heads and need to use a good randomization device (*like an app, or telegram sticker*)
- ▶ RPS becomes a fair game of luck

# Nash Theorem

- ▶ **Nash Theorem:** In any *finite game*, at least one Nash equilibrium always exists (potentially in mixed strategies).
- ▶ A **finite game** is a game with a finite number of players, each with a finite number of pure strategies.

## Theory: Revisiting **Dominance**

- ▶ For player 1, a pure strategy  $s_1$  is **strictly dominated** by a mixed strategy  $p_1$  if

$$EU_1(p_1, s_2) > u_1(s_1, s_2) \quad \text{for all } s_2 \in S_2$$

- ▶ For player 1, a pure strategy  $s_1$  is **weakly dominated** by a mixed strategy  $p_1$  if

$$EU_1(p_1, s_2) \geq u_1(s_1, s_2) \quad \text{for all } s_2 \in S_2$$

$$EU_1(p_1, s_2) > u_1(s_1, s_2) \quad \text{for some } s_2 \in S_2$$

- ▶ We can extend IESDS and IEWDS to allow for mixed strategies, i.e. at each stage, **eliminate pure strategies** that are dominated by some mixed strategies.



## Example of Iterated Elimination using Mixed Strategy

|          |   | Player II |      |
|----------|---|-----------|------|
|          |   | L         | R    |
| Player I | T | 1, 2      | 4, 1 |
|          | M | 4, 2      | 1, 1 |
|          | B | 2, 1      | 2, 3 |

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1. For P1, B is strictly dominated by  $p_1 = (0.5, 0.5, 0)$  (50%-50% mixture of T and M) since

$$EU_1(p_1, s_2) > u_1(B, s_2) \quad \forall s_2$$

## Example of Iterated Elimination using Mixed Strategy

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2. For P2, R is strictly dominated by L
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## Skill checklist for Part 2a

1. Specify all components of a game
2. Find strictly and weakly dominated strategies.
3. Identify strategies that can be eliminated by IESDS and IEWDS.
4. Find Nash equilibria in pure strategies.
5. Calculate payoffs under mixed strategies